



## STAINLESS STEEL SEAMLESS & WELDED PIPES CHEMICAL & PHYSICAL PROPERTIES

| Chemical Composition |           |          |            |          |          |            |           |            |                   | Tensile Test                                     |                                                      |                                                                                                                                                                                                                                                                                                                  |            |            |
|----------------------|-----------|----------|------------|----------|----------|------------|-----------|------------|-------------------|--------------------------------------------------|------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|------------|
| Grade                | C         | Si       | Mn         | P<br>Max | S<br>Max | Cr         | Mo        | Ni         | Other<br>Elements | Tensile<br>Strength<br>kg/mm <sup>2</sup><br>min | Yield Point<br>Strength<br>kg/mm <sup>2</sup><br>min | Elongation in 2 or 50mm min%<br>Curved strip and Tubular Specimens                                                                                                                                                                                                                                               |            |            |
|                      |           |          |            |          |          |            |           |            |                   |                                                  |                                                      | Longitudinal                                                                                                                                                                                                                                                                                                     |            | Transverse |
|                      |           |          |            |          |          |            |           |            |                   |                                                  |                                                      | t ≤ 5/16in                                                                                                                                                                                                                                                                                                       | t ≥ 5/16in | t ≤ 5/16in |
| TP304                | 0.08 max  | 0.75 max | 2.0 max    | 0.040    | 0.030    | 18.0-20.0  |           | 8.00-11.0  | -                 | 52.52                                            | 20.90                                                | 35                                                                                                                                                                                                                                                                                                               | -          | 25         |
| TP304H               | 0.04-0.10 | 0.75 max | 2.0 max    | 0.040    | 0.030    | 18.0-20.0  |           | 8.00-11.0  | -                 | 52.52                                            | 20.90                                                | 35                                                                                                                                                                                                                                                                                                               | -          | 25         |
| TP304L               | 0.035 max | 0.75 max | 2.0 max    | 0.040    | 0.030    | 18.0-20.0  |           | 8.00-13.0  | -                 | 49.46                                            | 17.34                                                | 35                                                                                                                                                                                                                                                                                                               | -          | 25         |
| TP304N               | 0.08 max  | 0.75 max | 2.0 max    | 0.040    | 0.030    | 18.0-20.0  |           | 8.00-11.0  | N 0.10-0.16       | 56.10                                            | 24.61                                                | 35                                                                                                                                                                                                                                                                                                               | -          | 25         |
| TP304LN              | 0.035 max | 0.75 max | 2.0 max    | 0.040    | 0.030    | 18.0-20.0  |           | 8.00-11.0  | N 0.10-0.16       | 52.52                                            | 20.90                                                | 35                                                                                                                                                                                                                                                                                                               | -          | 25         |
| TP309S               | 0.08 max  | 0.75 max | 2.0 max    | 0.045    | 0.030    | 22.0-24.0  | 0.75 Max  | 12.0-15.0  | -                 | 52.52                                            | 20.90                                                | 35                                                                                                                                                                                                                                                                                                               | -          | 25         |
| TP310S               | 0.08 max  | 0.75 max | 2.0 max    | 0.045    | 0.030    | 24.0-26.0  | 0.75 Max  | 19.0-22.0  | -                 | 52.52                                            | 20.90                                                | 35                                                                                                                                                                                                                                                                                                               | -          | 25         |
| TP316                | 0.08 max  | 0.75 max | 2.0 max    | 0.040    | 0.030    | 16.0-18.0  | 2.00-3.00 | 11.0-14.0  | -                 | 52.52                                            | 20.90                                                | 35                                                                                                                                                                                                                                                                                                               | -          | 25         |
| TP316H               | 0.04-0.10 | 0.75 max | 2.0 max    | 0.040    | 0.030    | 16.0-18.0  | 2.00-3.00 | 11.0-14.0  | -                 | 52.52                                            | 20.90                                                | 35                                                                                                                                                                                                                                                                                                               | -          | 25         |
| TP316L               | 0.035 max | 0.75 max | 2.0 max    | 0.040    | 0.030    | 16.0-18.0  | 2.00-3.00 | 10.0-15.0  | -                 | 49.46                                            | 17.34                                                | 35                                                                                                                                                                                                                                                                                                               | -          | 25         |
| TP316N               | 0.08 max  | 0.75 max | 2.0 max    | 0.040    | 0.030    | 16.0-18.0  | 2.00-3.00 | 11.0-14.0  | N 0.10~0.16       | 56.10                                            | 24.61                                                | 35                                                                                                                                                                                                                                                                                                               | -          | 25         |
| TP316LN              | 0.035 max | 0.75 max | 2.0 max    | 0.040    | 0.030    | 16.0-18.0  | 2.00~3.00 | 11.0-14.0  | N 0.10~0.16       | 52.52                                            | 20.90                                                | 35                                                                                                                                                                                                                                                                                                               | -          | 25         |
| TP317                | 0.08 max  | 0.75 max | 2.0 max    | 0.040    | 0.030    | 18.0-20.0  | 3.00~4.00 | 11.0-14.0  | -                 | 52.52                                            | 20.90                                                | 35                                                                                                                                                                                                                                                                                                               | -          | 25         |
| TP317L               | 0.035 max | 0.75 max | 2.0 max    | 0.040    | 0.030    | 18.0-20.0  | 3.00~4.00 | 11.0-15.0  | -                 | 52.52                                            | 20.90                                                | 35                                                                                                                                                                                                                                                                                                               | -          | 25         |
| TP321                | 0.08 max  | 0.75 max | 2.0 max    | 0.040    | 0.030    | 17.0-20.0  |           | 9.0-13.0   | Ti5xC%-0.70       | 52.52                                            | 20.90                                                | 35                                                                                                                                                                                                                                                                                                               | -          | 25         |
| TP321H               | 0.04-0.10 | 0.75 max | 2.0 max    | 0.040    | 0.030    | 17.0-20.0  |           | 9.0-13.0   | Ti4xC%-0.60       | 52.52                                            | 20.90                                                | 35                                                                                                                                                                                                                                                                                                               | -          | 25         |
| TP347                | 0.08 max  | 0.75 max | 2.0 max    | 0.040    | 0.030    | 17.0-20.0  |           | 9.0-13.0   | Nb+TA10xC%-1.00   | 52.52                                            | 20.90                                                | 35                                                                                                                                                                                                                                                                                                               | -          | 25         |
| TP347H               | 0.04-0.10 | 0.75 max | 2.0 max    | 0.040    | 0.030    | 17.0-20.0  |           | 9.0-13.0   | Nb+TA8xc%~1       | 52.52                                            | 20.90                                                | 35                                                                                                                                                                                                                                                                                                               | -          | 25         |
| TP348                | 0.08 max  | 0.75 max | 2.0 max    | 0.040    | 0.030    | 17.0-20.0  |           | 9.0-13.0   | Nb+TA10xc%~1      | 52.52                                            | 20.90                                                | 35                                                                                                                                                                                                                                                                                                               | -          | 25         |
| TP348H               | 0.04-0.10 | 0.75 max | 2.0 max    | 0.040    | 0.030    | 17.0-20.0  |           | 9.0-13.0   | Nb+TA8xc%~1       | 52.52                                            | 20.90                                                | 35                                                                                                                                                                                                                                                                                                               | -          | 25         |
| TPXM-10              | 0.08 max  | 1.00 max | 8.00-10.00 | 0.040    | 0.030    | 19.0-21.50 |           | 5.50-7.50  | N0.15~0.40        | 63.22                                            | 35.18                                                | 35                                                                                                                                                                                                                                                                                                               | -          | 25         |
| TPXM-11              | 0.04 max  | 1.00 max | 8.00-10.00 | 0.040    | 0.030    | 19.0-21.50 |           | 5.50-7.50  | N0.15~0.40        | 63.22                                            | 35.18                                                | 35                                                                                                                                                                                                                                                                                                               | -          | 25         |
| A240 TP304           | 0.08 max  | 0.75 max | 2.0 max    | 0.045    | 0.030    | 18.0-20.0  | -         | 8.00-10.50 | N0.10 Max         |                                                  |                                                      | Class 1 :<br>Double welded pipes & full<br>Radiography<br><br>Class 2 :<br>Double welded no Radiography<br><br>Class 3 :<br>Single welded full Radiography<br><br>Class 4 :<br>Single welded full Radiography<br>rootpass without addition of<br>filler metal<br><br>Class 5 :<br>Double welded spot Radiography |            |            |
| TP310S               | 0.08 max  | 0.75 max | 2.0 max    | 0.045    | 0.030    | 24.0-26.0  | -         | 19.0-22.0  |                   |                                                  |                                                      |                                                                                                                                                                                                                                                                                                                  |            |            |
| TP316                | 0.08 max  | 0.75 max | 2.0 max    | 0.045    | 0.030    | 16.0-18.0  | 2.0-3.0   | 10.0-14.0  | N0.10 Max         |                                                  |                                                      |                                                                                                                                                                                                                                                                                                                  |            |            |
| TP316L               | 0.035 max | 0.75 max | 2.0 max    | 0.045    | 0.030    | 16.0-18.0  | 2.0-3.0   | 10.0-14.0  | N0.10 Max         |                                                  |                                                      |                                                                                                                                                                                                                                                                                                                  |            |            |
| TP317L               | 0.035 max | 0.75 max | 2.0 max    | 0.045    | 0.030    | 18.0-20.0  | 3.0-4.0   | 11.0-15.0  | N0.10 Max         |                                                  |                                                      |                                                                                                                                                                                                                                                                                                                  |            |            |
| TP321                | 0.08 max  | 0.75 max | 2.0 max    | 0.045    | 0.030    | 17.0-19.0  | -         | 9.0-12.0   | N0.10 Max         |                                                  |                                                      |                                                                                                                                                                                                                                                                                                                  |            |            |
| TP347                | 0.08 max  | 0.75 max | 2.0 max    | 0.045    | 0.030    | 17.0-19.0  | -         | 9.0-13.0   | Cb+TA10xC/1.0     |                                                  |                                                      |                                                                                                                                                                                                                                                                                                                  |            |            |